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Optimization and Algorithms in Sparse Regression

Cramér Mini-Conference 2025

Johan Larsson

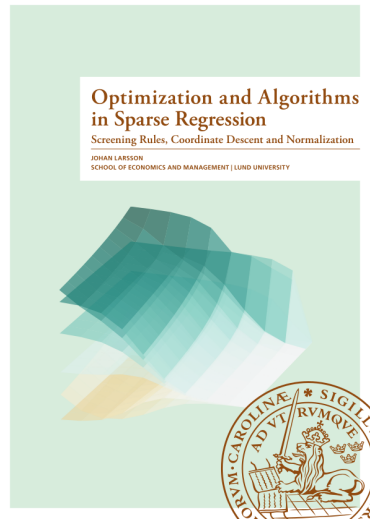
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March 24, 2025

Overarching Theme

Practical concerns in sparse regression:
speed, accuracy, and reproducibility.



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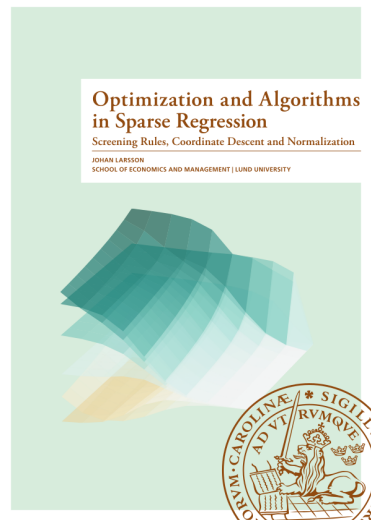
Overview

I–III Screening rules

IV Benchmarking optimization methods

V Coordinate descent (an optimization method) for SLOPE

VI Normalization and regularization



Screening Rules (Papers I–III)

Motivation

Fitting models to sets of millions or billions of features is computationally costly, and exacerbated by the fact we often need to do cross-validation.

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Basic Idea of Screening

- The support set is small in sparse regression, especially when $p \gg n$.¹
- If we could estimate the model's support **before** fitting the model, then we would save a lot of time.

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General Outline

1. Estimate how likely it is that a feature is in the support set.
2. If unlikely, discard it and fit a smaller model.
3. If we are wrong, just refit the model with missing features added.

¹The lasso can for instance at most select $\min(n, p)$ features.

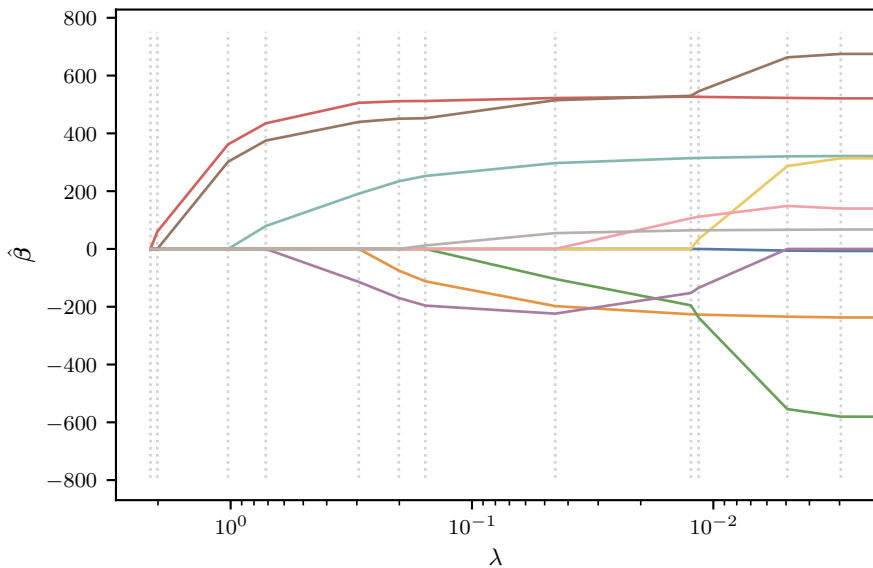


Figure 1: The regularization path for the lasso. The path is piecewise linear.

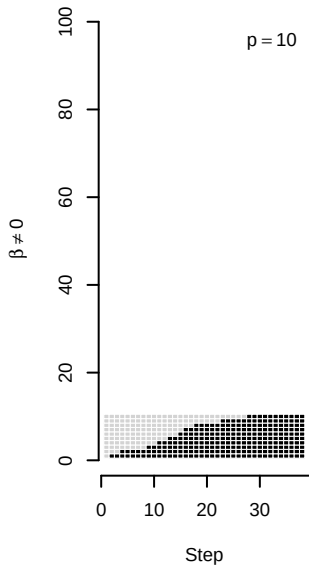


Figure 2: Non-zero coefficients along the path, ordered by the time they enter the model. The number of observations, n , is 50.

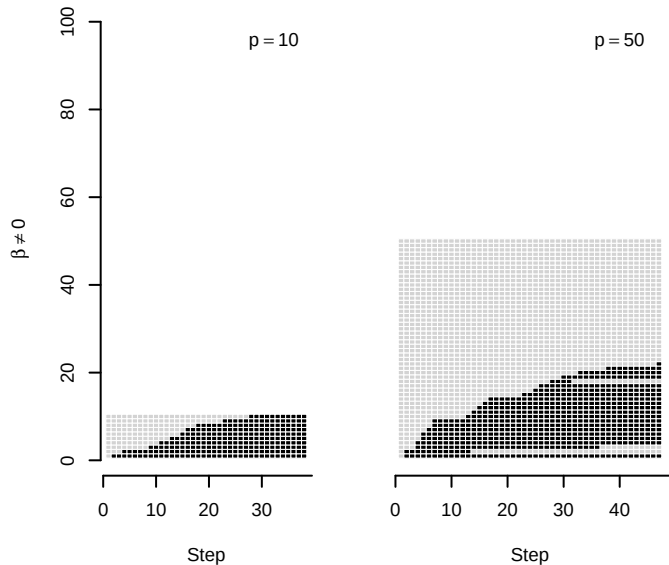


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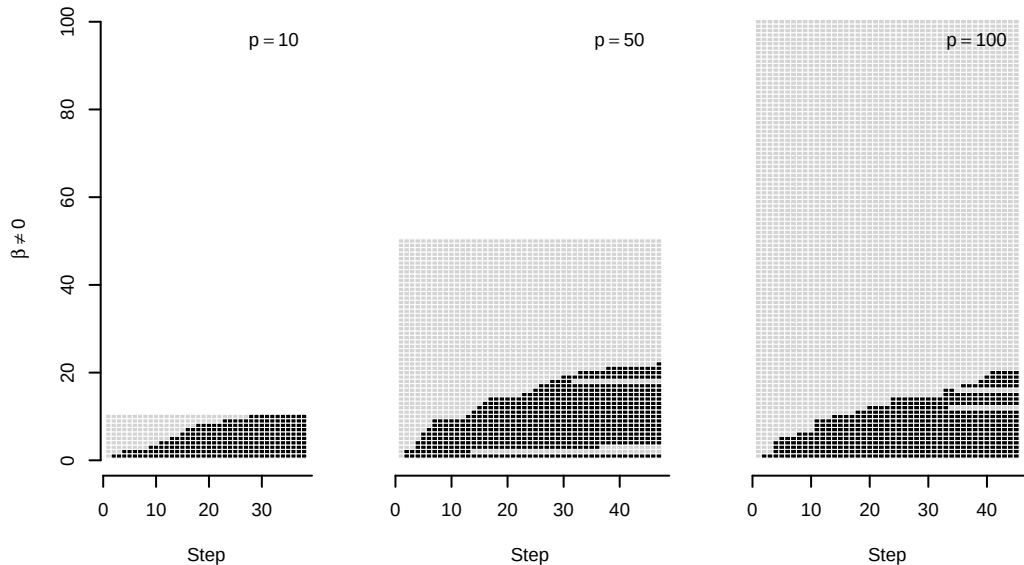


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Optimality Condition

β is optimal if $\mathbf{0} \in \nabla g(\beta) + \partial h(\beta)$, where $\partial h(\beta)$ is the subdifferential of the penalty term h .

Optimality Conditions and the Gradient

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Example: Lasso

$$\partial h(\beta)_j = \begin{cases} \{\lambda \operatorname{sign}(\beta_j)\} & \text{if } \beta_j \neq 0, \\ [-\lambda, \lambda] & \text{if } \beta_j = 0. \end{cases}$$

So if $|\nabla g(\hat{\beta})_j| < \lambda$, then $\hat{\beta}_j = 0$.

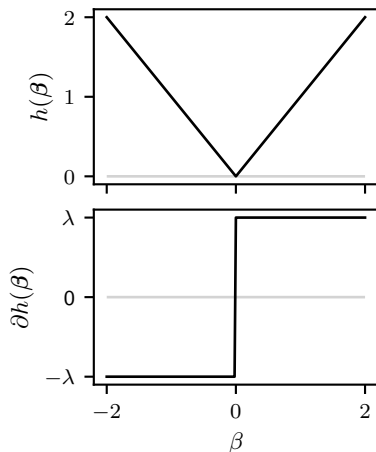


Figure 3: The lasso penalty $h(\beta) = \lambda \|\beta\|_1$ and its subdifferential at $\lambda = 1$.

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Screening Rule Template

1. Replace $\nabla g(\hat{\beta})$ with an estimate $\tilde{\nabla} g(\beta)$.
2. If $|\tilde{\nabla} g(\beta)_j| < \lambda$, discard feature j .

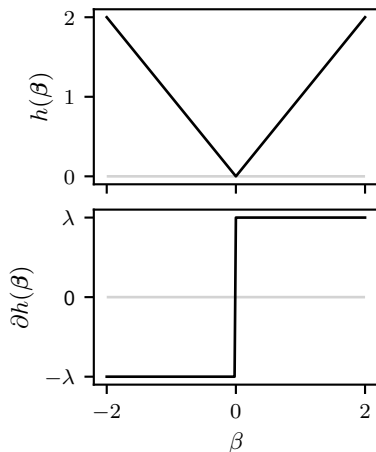


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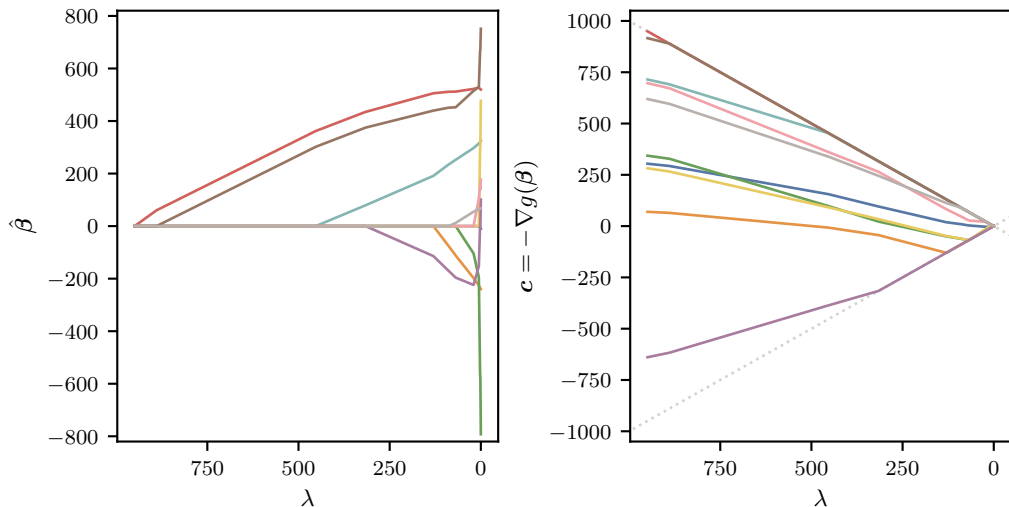


Figure 4: Coefficients and correlations (c) along the lasso path. If the gradient of any given coefficient is smaller than the penalty (dotted line), then the coefficient has to be zero.

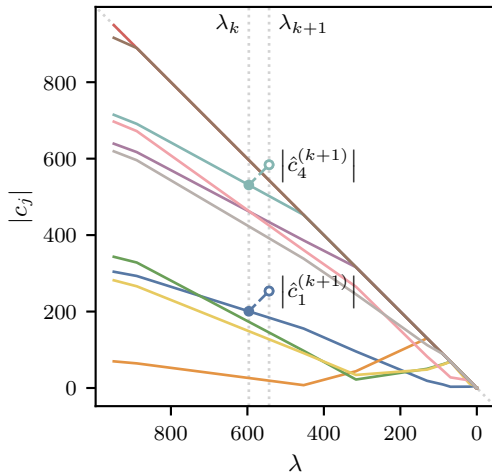


Figure 5: The strong rule for the lasso (Tibshirani et al. 2012). $|\hat{c}_j^{(k+1)}| = |c_j^{(k)}| + \lambda_k - \lambda_{k+1}$ is the absolute correlation estimate for feature j at step $k + 1$.

Strong Rule for SLOPE (Paper I)

Same idea as for lasso strong rule! Just need the subdifferential.

In our paper², we derive the subdifferential for SLOPE/OWL and present an efficient algorithm for using it to screen features.



²Johan Larsson, Małgorzata Bogdan, and Jonas Wallin (Dec. 6–12, 2020). “The Strong Screening Rule for SLOPE”. In: *Advances in Neural Information Processing Systems* 33. 34th Conference on Neural Information Processing Systems (NeurIPS 2020). Ed. by Hugo Larochelle et al. Vol. 33. Virtual: Curran Associates, Inc., pp. 14592–14603. ISBN: 978-1-71382-954-6

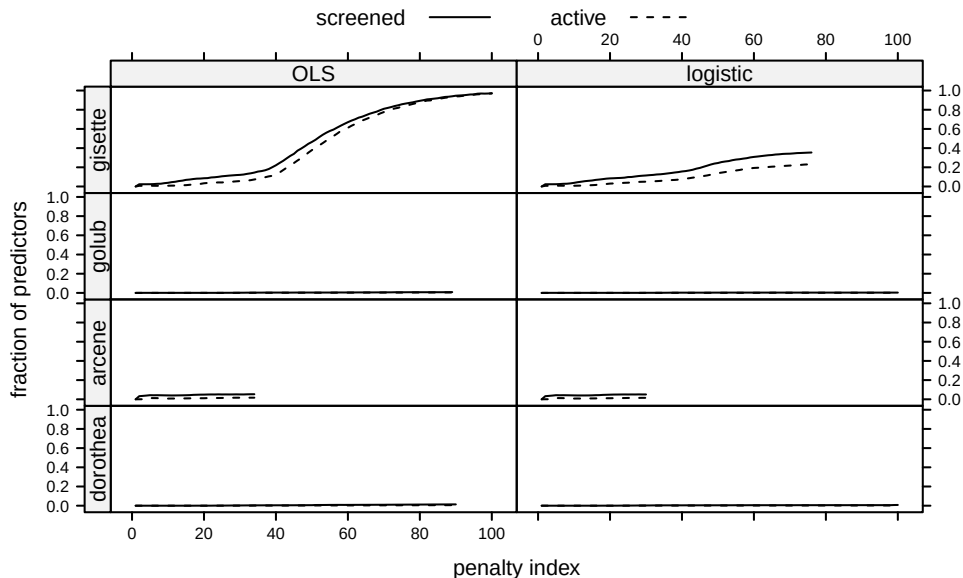


Figure 6: Effectiveness of the strong screening rule for SLOPE on some real data sets.

Table 1: Time to fit lasso paths using either the strong screening rule or no rule.

Dataset	Model	n	p	Time (s)	
				No Screening	Screening
dorothea	Logistic	800	88 119	914	14
e2006-tfidf	Least squares	3308	150 358	43 353	4944
news20	Multinomial	1000	62 061	5485	517
physician	Poisson	4406	25	34	34

Look-Ahead Screening (Paper II)

- There's nothing special about the current step's λ in a screening rule, so why not look further ahead?³

³Johan Larsson (Sept. 6, 2021). "Look-Ahead Screening Rules for the Lasso". In: *22nd European Young Statisticians Meeting - Proceedings*. 22nd European Young Statisticians Meeting. Ed. by Andreas Makridis et al. Athens, Greece: Panteion university of social and political sciences, pp. 61–65. ISBN: 978-960-7943-23-1

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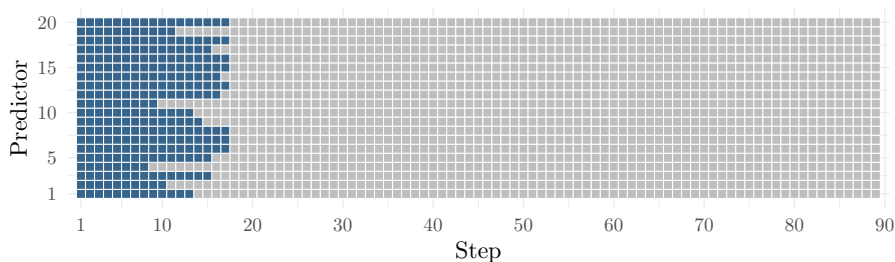


Figure 7: Lasso path for a sample of features from the leukemia data set. The squares show for which steps the feature can be discarded.

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The Hessian Screening Rule (Paper III)

A screening rule for the lasso that uses second-order information to obtain better estimates of future gradients.⁴

⁴Johan Larsson and Jonas Wallin (Nov. 28–Dec. 9, 2022). “The Hessian Screening Rule”. In: *Advances in Neural Information Processing Systems 35*. 36th Conference on Neural Information Processing Systems (NeurIPS 2022). Ed. by Sanmi Koyejo et al. Vol. 35. New Orleans, USA: Curran Associates, Inc., pp. 15823–15835. ISBN: 978-1-71387-108-8

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Improved Screening

Uses information about the gradient of the gradients along the path, which is less conservative than the strong rule.

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Better Warm Starts

Allows also to estimate the entire coefficient vector for the next step.

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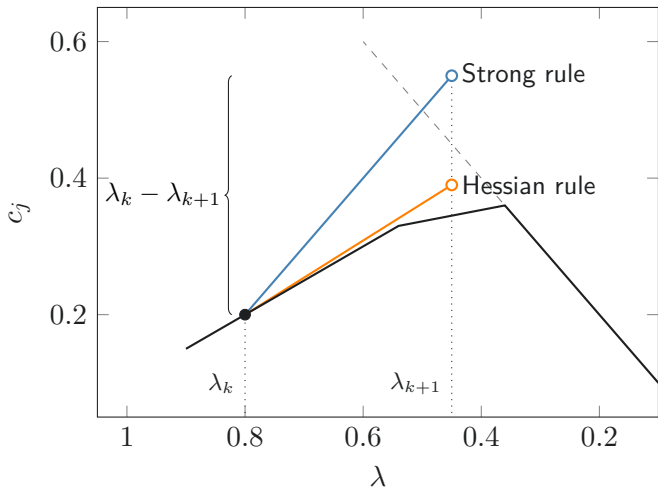


Figure 8: The strong and Hessian screening rules in action

Coordinate Descent for SLOPE (Paper V)

- SLOPE has appealing properties but algorithms for solving SLOPE are^a slow (relative to the lasso)

^aUsed to be!

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- Lasso solvers are fast because they use coordinate descent.

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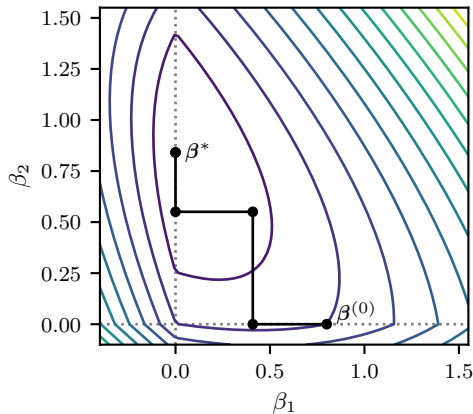


Figure 9: A basic example of coordinate descent in two dimensions

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- SLOPE has appealing properties but algorithms for solving SLOPE are^a slow (relative to the lasso)
- Lasso solvers are fast because they use coordinate descent.
- Unfortunately cannot use basic coordinate descent for SLOPE

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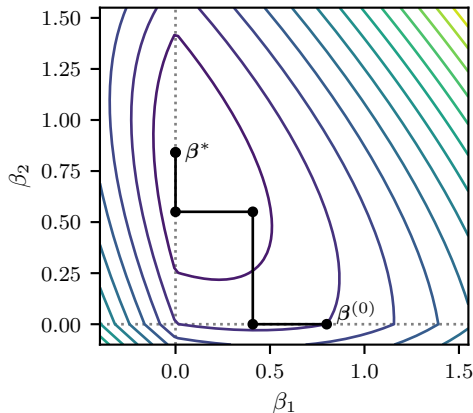


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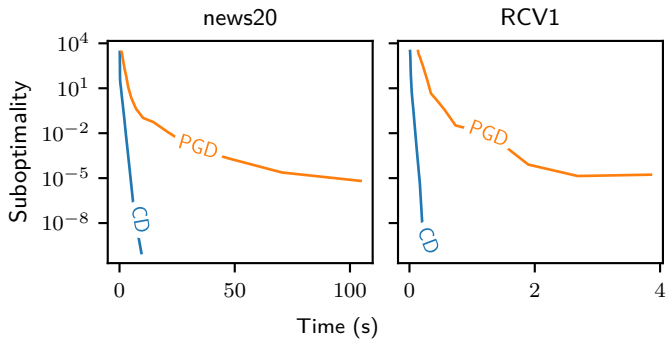


Figure 10: Coordinate descent versus proximal gradient descent for the lasso.

Coordinate Descent and Separability

Problem

Cannot use basic coordinate descent for SLOPE since the sorted ℓ_1 norm is **inseparable**:

$$h(\beta) = \sum_{j=1}^p \lambda_j |\beta_{(j)}|.$$

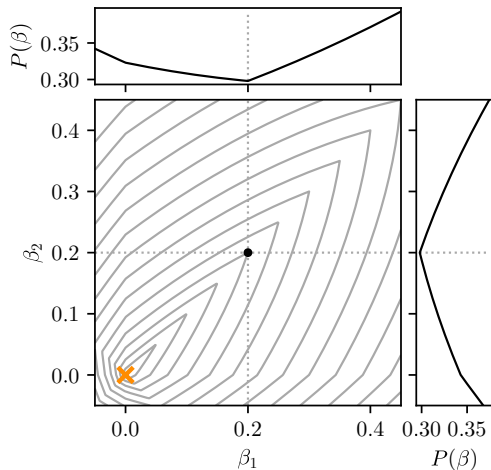


Figure 11: A *naive* coordinate descent algorithm cannot advance from the current iterate (●) to reach the optimum (×).

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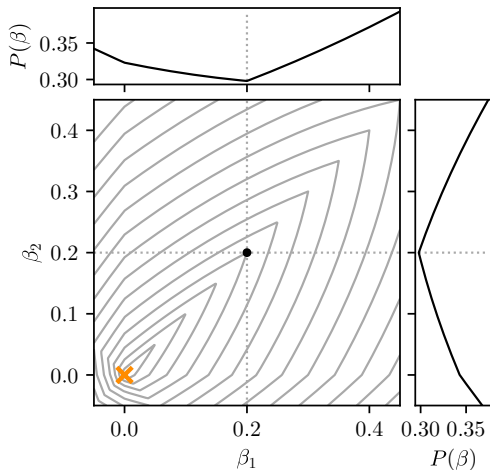


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Idea

Alternate between gradient descent and coordinate descent **on the clusters**.

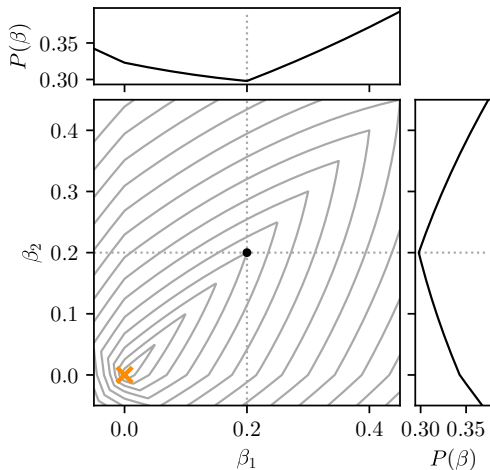


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Our algorithm is a combination of coordinate descent and proximal gradient descent:⁵

- Every v th iteration, take a full proximal gradient step. This allows clusters to split (or merge).
- At all other iterations, take coordinate descent steps on the clusters.

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Hybrid Algorithm

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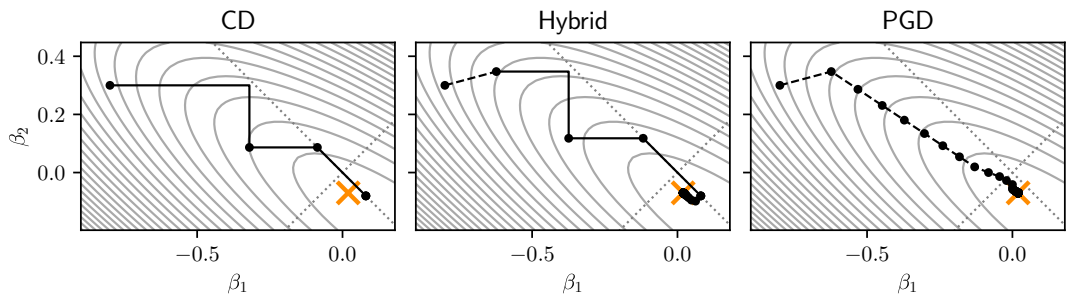


Figure 12: Our algorithm (hybrid) is a combination of CD and PGD.

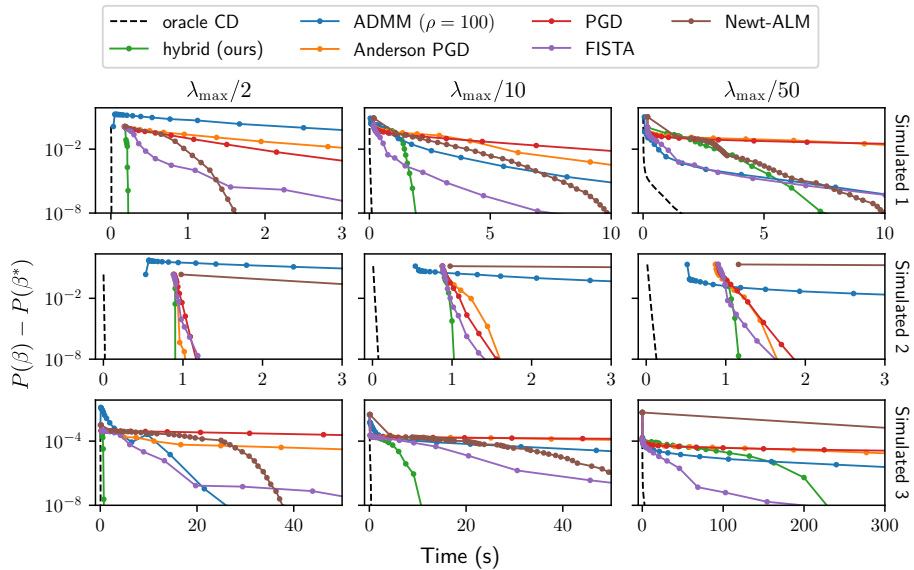
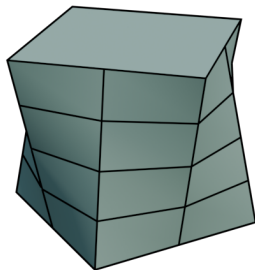


Figure 13: Benchmarks on simulated data. Scenario 1: $n = 200$ and $p = 20\,000$, X . Scenario 2: $n = 20\,000$ and $p = 200$. Scenario 3: $n = 200$, $p = 200\,000$, and sparse X .

The SLOPE Package(s)

LibSLOPE

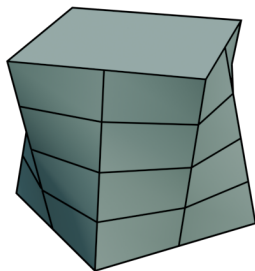
- C++ library for solving SLOPE using the hybrid coordinate method
- Parallelized fitting and cross-validation
- Algorithmic improvements to SLOPE thresholding and cluster handling for sparse data



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Packages

Language	Package	URL
R	SLOPE	https://jolars.github.io/SLOPE
Python	sortedl1	https://jolars.github.io/sortedl1
C++	libslope	https://jolars.github.io/libslope

The Choice of Normalization Affects Shrinkage in Regularized Regression

The Elastic Net

Linear regression plus a combination of the ℓ_1 and ℓ_2 penalties:

$$\hat{\boldsymbol{\beta}} = \arg \min_{\boldsymbol{\beta} \in \mathbb{R}^p} \left(\frac{1}{2} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \underbrace{\lambda_1 \|\boldsymbol{\beta}\|_1}_{\text{lasso}} + \underbrace{\frac{\lambda_2}{2} \|\boldsymbol{\beta}\|_2^2}_{\text{ridge}} \right)$$

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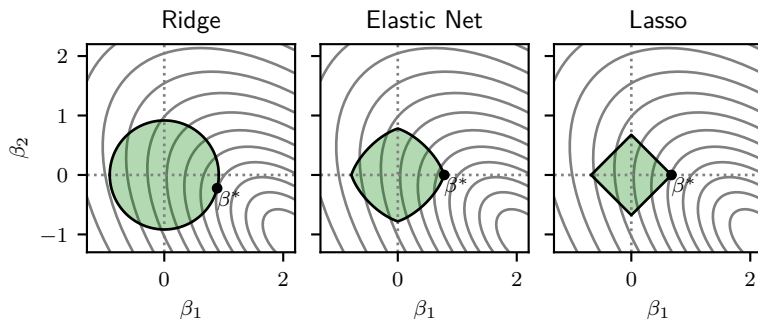


Figure 14: Elastic net is a combination of the lasso and ridge.

Scale Sensitivity

Lasso and ridge penalties are **norms**—feature scales matter!

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Example

Assume

$$\mathbf{X} \sim \text{Normal} \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix} \right), \quad \boldsymbol{\beta}^* = \begin{bmatrix} \frac{1}{2} \\ 1 \end{bmatrix}.$$

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Model	$\hat{\boldsymbol{\beta}}$	$\hat{\boldsymbol{\beta}}_{\text{std}}$
OLS	$\begin{bmatrix} 0.50 & 1.00 \end{bmatrix}^{\top}$	$\begin{bmatrix} 1.00 & 1.00 \end{bmatrix}^{\top}$

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Large scale means less penalization because the size of β_j can be smaller for an equivalent effect (on $\mathbf{y}$).

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- After fitting, we transform the coefficients back to their original scale via

$$\hat{\beta}_j = \frac{\hat{\beta}_j^{(n)}}{s_j} \quad \text{for } j = 1, 2, \dots, p,$$

where $\hat{\beta}_j^{(n)}$ is a coefficient from the normalized problem.

Table 2: Common ways to normalize $\mathbf{X}$

Normalization	Centering (c_j)	Scaling (s_j)
Standardization	$\bar{x}_j = \frac{1}{n} \sum_{i=1}^n x_{ij}$	$\sqrt{\frac{1}{n} \sum_{i=1}^n (x_{ij} - \bar{x}_j)^2}$
Min–Max	$\min_i(x_{ij})$	$\max_i(x_{ij}) - \min_i(x_{ij})$
Unit Vector (L2)	0	$\sqrt{\sum_{i=1}^n x_{ij}^2}$
Max–Abs	0	$\max_i(x_{ij})$
Adaptive Lasso	0	β_j^{OLS}

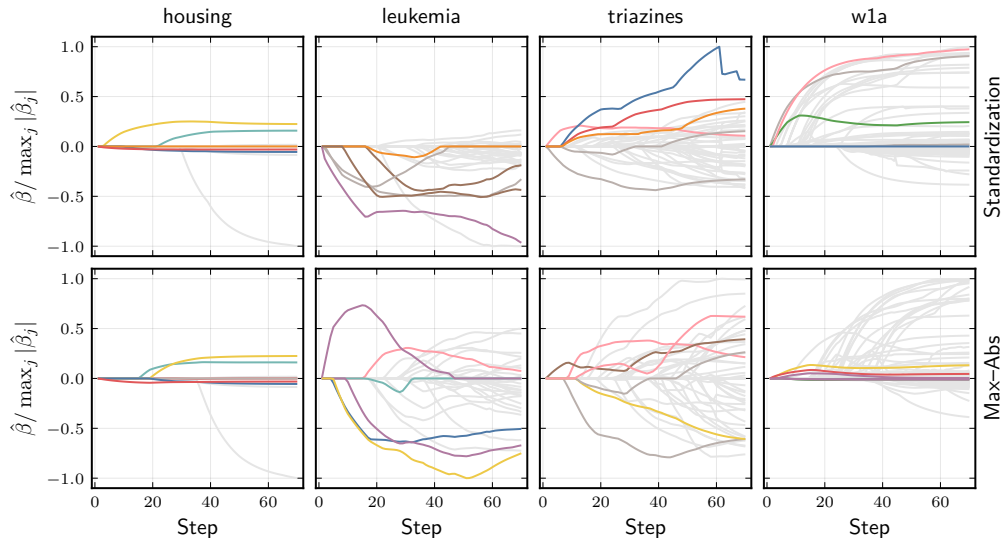


Figure 15: Normalization matters. Lasso paths under two different types of normalization (standardization and max-abs normalization). The union of the first ten features selected in any of the settings are colored.

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- Documentation for popular software advocate different normalization strategies when data is sparse.
- Approach depends on field: statisticians standardize, signal processors use ℓ_1 normalization, machine learners scale to $[0, 1]$ or $[-1, 1]$.)

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- Documentation for popular software advocate different normalization strategies when data is sparse.
- Approach depends on field: statisticians standardize, signal processors use ℓ_1 normalization, machine learners scale to $[0, 1]$ or $[-1, 1]$.)

Motivation

- Normalization matters, yet there is almost no research on this topic.
- Everyone agrees you need to normalize, but how to do so is usually motivated by being “standard”.
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Our Paper: Focus on Binary Features

In our paper, we focus on binary features particularly with respect to **class balance**.

Orthogonal Features

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$$\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}} = \text{diag}(\tilde{\mathbf{x}}_1^\top \tilde{\mathbf{x}}_1, \dots, \tilde{\mathbf{x}}_p^\top \tilde{\mathbf{x}}_p),$$

then there is:⁶:

$$\hat{\beta}_j = \frac{S_{\lambda_1}(\tilde{\mathbf{x}}_j^\top \mathbf{y})}{s_j(\tilde{\mathbf{x}}_j^\top \tilde{\mathbf{x}}_j + \lambda_2)},$$

where

$$S_\lambda(z) = \text{sign}(z) \max(|z| - \lambda, 0).$$

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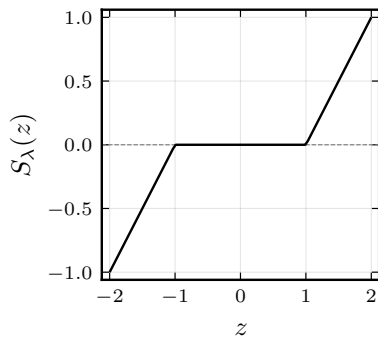


Figure 16: Soft thresholding

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Noiseless Case for Binary Features

In the noiseless case, we have

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which we will rely on for the sequel.

- Indicates there might be no (simple) s_j that will work for the elastic net.

Parameterization

We introduce a scaling parameterization

$$s_j = (q - q^2)^\delta, \quad \delta \geq 0.$$

so that $\delta = 0$ corresponds to no scaling,
 $\delta = 1/2$ standardization, and $\delta = 1$ to
variance scaling.

⁷Johan Larsson and Jonas Wallin (Jan. 21, 2025). *The Choice of Normalization Influences Shrinkage in Regularized Regression*. DOI: [10.48550/arXiv.2501.03821](https://doi.org/10.48550/arXiv.2501.03821). arXiv: [2501.03821](https://arxiv.org/abs/2501.03821) [stat]. URL: <http://arxiv.org/abs/2501.03821> (visited on 01/22/2025). Pre-published

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Mixed Data (Not Covered Here)

We consider a mix of binary and continuous features.

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Probability of Selection

Assuming $\mathbf{X}$ is fixed and noise is normal, we can compute the probability of selection.

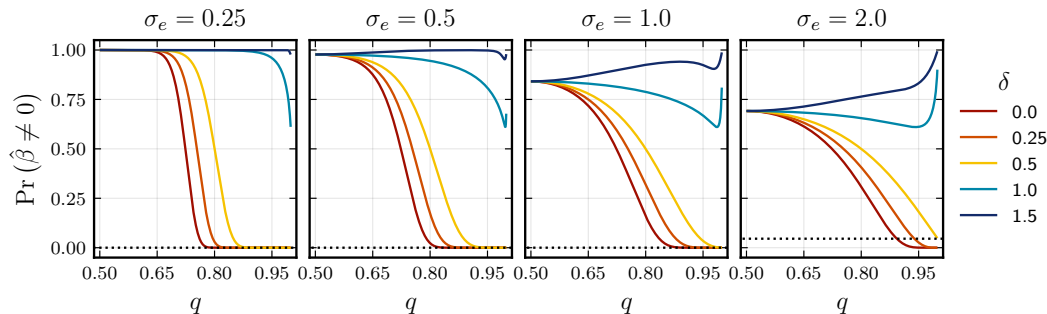


Figure 17: Probability that the elastic net selects a feature across different noise levels (σ_ϵ), types of normalization (δ), and class balance (q). The dashed line is asymptotic behavior for $\delta = 1/2$. Scaling used is $s_j \propto (q - q^2)^\delta$.

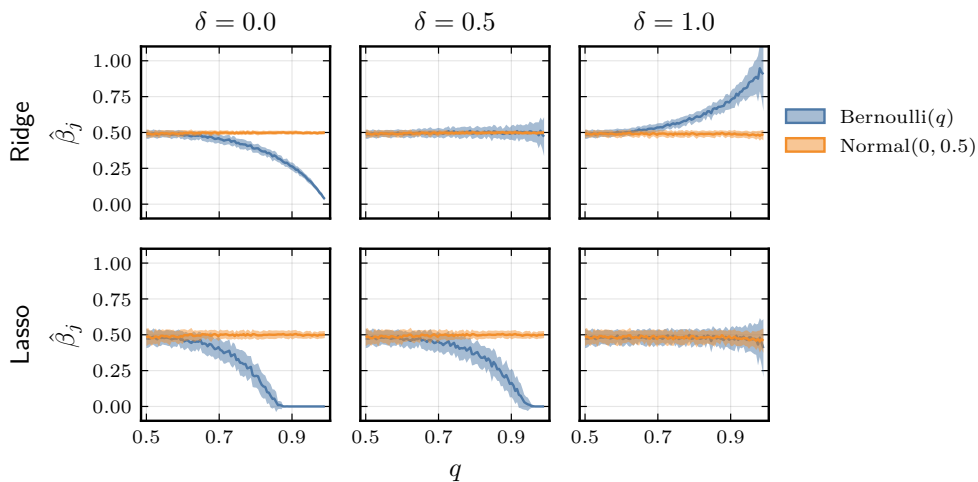


Figure 18: Comparison between lasso and ridge estimators for features generated to resemble features from various distributions.

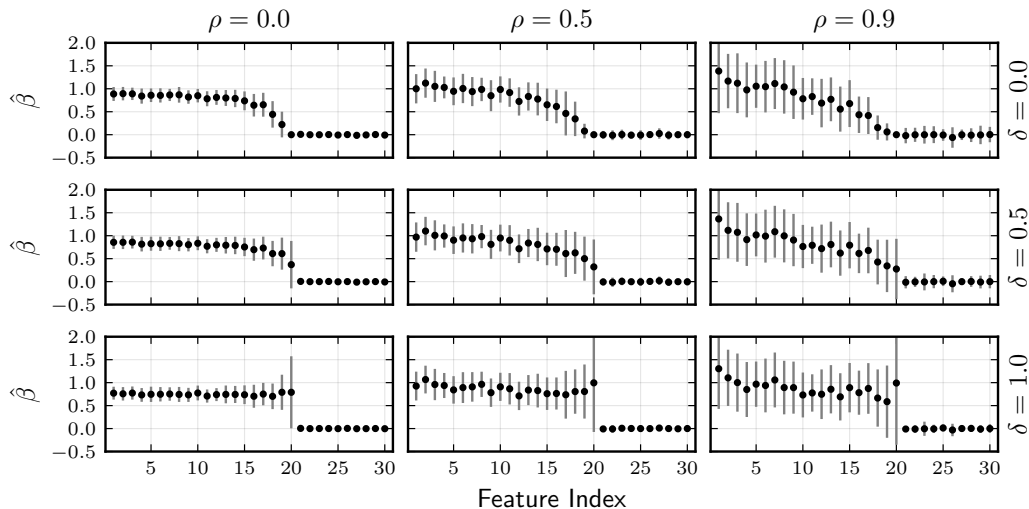


Figure 19: Lasso estimates for first 30 coefficients. First 20 features are true signals with a geometrically decreasing class balance from 0.5 to 0.99. ρ is a measure of autocorrelation.

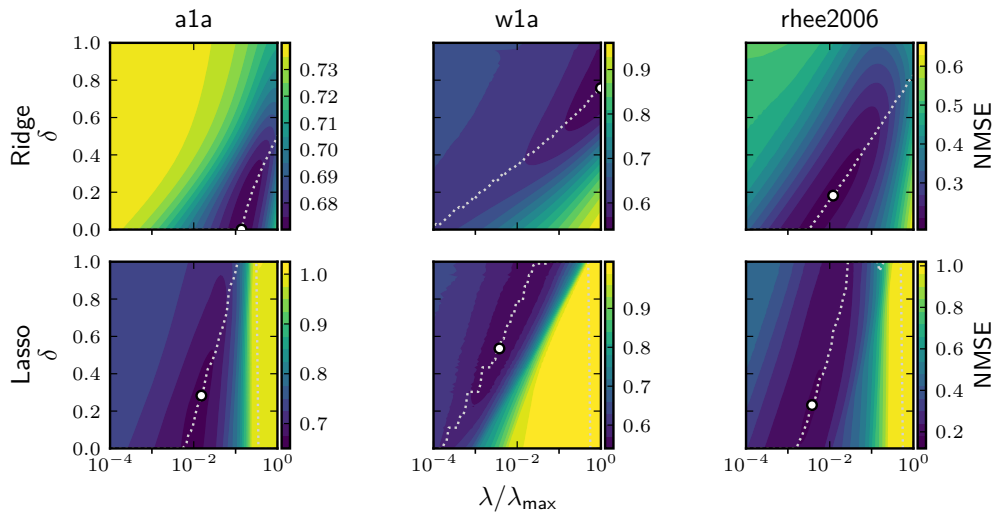


Figure 20: Contour plots of hold-out (validation set) error across a grid of δ and λ values for the lasso and ridge.

Screening Rules (I–III)

Screening rules for SLOPE and the lasso.

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Benchopt (IV)

Benchmarking optimization methods.

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A hybrid coordinate descent method for SLOPE.

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Normalization and Regularization (VI)

The interplay between normalization and regularization.

Thank you!

-  Larsson, Johan (Sept. 6, 2021). “Look-Ahead Screening Rules for the Lasso”. In: *22nd European Young Statisticians Meeting - Proceedings*. 22nd European Young Statisticians Meeting. Ed. by Andreas Makridis et al. Athens, Greece: Panteion university of social and political sciences, pp. 61–65. ISBN: 978-960-7943-23-1.
-  Larsson, Johan, Małgorzata Bogdan, and Jonas Wallin (Dec. 6–12, 2020). “The Strong Screening Rule for SLOPE”. In: *Advances in Neural Information Processing Systems 33*. 34th Conference on Neural Information Processing Systems (NeurIPS 2020). Ed. by Hugo Larochelle et al. Vol. 33. Virtual: Curran Associates, Inc., pp. 14592–14603. ISBN: 978-1-71382-954-6.
-  Larsson, Johan, Quentin Klopfenstein, et al. (Apr. 25–27, 2023). “Coordinate Descent for SLOPE”. In: *Proceedings of the 26th International Conference on Artificial Intelligence and Statistics*. AISTATS 2023. Ed. by Francisco Ruiz, Jennifer Dy, and Jan-Willem van de Meent. Vol. 206. Proceedings of Machine Learning Research. Valencia, Spain: PMLR, pp. 4802–4821.

-  Larsson, Johan and Jonas Wallin (Nov. 28–Dec. 9, 2022). “The Hessian Screening Rule”. In: *Advances in Neural Information Processing Systems 35*. 36th Conference on Neural Information Processing Systems (NeurIPS 2022). Ed. by Sanmi Koyejo et al. Vol. 35. New Orleans, USA: Curran Associates, Inc., pp. 15823–15835. ISBN: 978-1-71387-108-8.
-  — (Jan. 21, 2025). *The Choice of Normalization Influences Shrinkage in Regularized Regression*. DOI: [10.48550/arXiv.2501.03821](https://doi.org/10.48550/arXiv.2501.03821). arXiv: [2501.03821 \[stat\]](https://arxiv.org/abs/2501.03821). URL: <http://arxiv.org/abs/2501.03821> (visited on 01/22/2025). Pre-published.
-  Tibshirani, Robert et al. (Mar. 2012). “Strong Rules for Discarding Predictors in Lasso-Type Problems”. In: *The Journal of the Royal Statistical Society, Series B (Statistical Methodology)* 74.2, pp. 245–266. ISSN: 1369-7412. DOI: [10/c4bb85](https://doi.org/10/c4bb85).

Paper I Extras

Strong Rule Algorithm for SLOPE

Algorithm 1: Basic algorithm for the strong rule for SLOPE

Data: $\mathbf{c} \in \mathbb{R}^p$, $\boldsymbol{\lambda} \in \mathbb{R}^p$, where $\lambda_1 \geq \dots \geq \lambda_p \geq 0$

$\mathcal{S}, \mathcal{B} \leftarrow \emptyset$;

for $i \leftarrow 1$ **to** p **do**

$\mathcal{B} \leftarrow \mathcal{B} \cup \{i\}$;

if $\sum_{j \in \mathcal{B}} (c_j - \lambda_j) \geq 0$ **then**

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Set

$$\mathbf{c} = |\nabla g(\hat{\boldsymbol{\beta}}) + \boldsymbol{\lambda}^{(k-1)} - \boldsymbol{\lambda}^{(k)}|_{\downarrow} \quad \boldsymbol{\lambda} = \boldsymbol{\lambda}^{(k)},$$

and run the algorithm above; the result is the predicted support for $\hat{\boldsymbol{\beta}}(\boldsymbol{\lambda}^{(k)})$ (subject to a permutation).

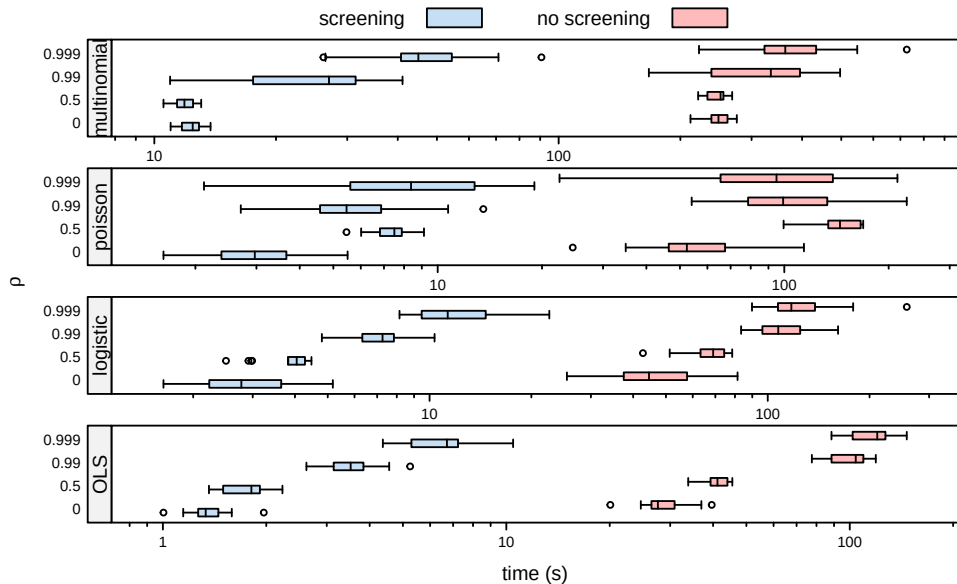


Figure 21: Time to fit a full SLOPE path with and without the strong rule

Paper II Extras

The Dual

Complementary to the primal problem. For the lasso, it is

$$\underset{\boldsymbol{\theta} \in \mathbb{R}^n}{\text{maximize}} \left(D(\boldsymbol{\theta}; \lambda) = \frac{1}{2} \mathbf{y}^\top \mathbf{y} - \frac{\lambda^2}{2} \left\| \boldsymbol{\theta} - \frac{\mathbf{y}}{\lambda} \right\|_2^2 \right)$$

The Dual

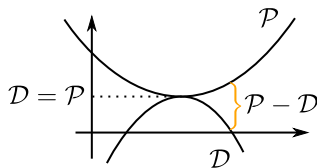
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Duality Gap

Difference between the primal and dual objectives. Tight at the optimum (for the lasso):

$$P(\boldsymbol{\beta}^*; \lambda) - D(\boldsymbol{\theta}^*; \lambda) = 0.$$



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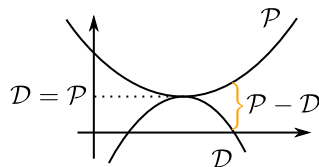
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Dual-Primal Relationship

The two problems are related via

$$\frac{\mathbf{y} - \mathbf{X}\boldsymbol{\beta}}{\lambda} = \boldsymbol{\theta}.$$

Inequality for Gaps Safe screening rule is **quadratic** in λ , which means we can find the next critical point easily:

$$\lambda_+ = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

where

$$a = (1 - |\mathbf{x}_j^\top \boldsymbol{\theta}_\lambda|)^2 - \frac{1}{2} \boldsymbol{\theta}_\lambda^\top \boldsymbol{\theta}_\lambda \|\mathbf{x}_j\|_2^2,$$

$$b = (\boldsymbol{\theta}_\lambda^\top \mathbf{y} - \|\boldsymbol{\beta}_\lambda\|_1) \|\mathbf{x}_j\|_2^2,$$

$$c = -\frac{1}{2} \|\mathbf{y} - \mathbf{X} \boldsymbol{\beta}_\lambda\|_2^2 \|\mathbf{x}_j\|_2^2.$$

Look-Ahead Screening

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$$c = -\frac{1}{2} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}_\lambda\|_2^2 \|\mathbf{x}_j\|_2^2.$$

Allows us to screen predictors for **all** upcoming steps

Paper III Extras

The availability of the Hessian inverse enables a better warm start:

$$\hat{\beta}(\lambda_{k+1})_{\mathcal{A}} = \hat{\beta}(\lambda_k)_{\mathcal{A}} + (\lambda_k - \lambda_{k+1})(X_{\mathcal{A}}^T X_{\mathcal{A}})^{-1} \text{sign}(\hat{\beta}(\lambda_k)_{\mathcal{A}})$$

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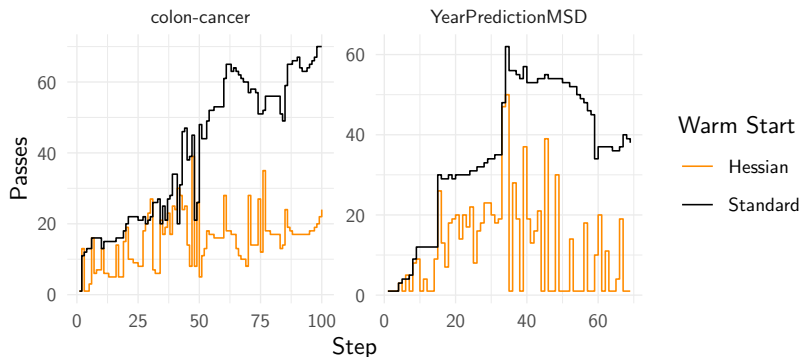


Figure 22: Number of passes of coordinate descent for two datasets using either Hessian warm starts or standard warm starts.

Updating the Hessian

Computing the Hessian and its inverse naively is expensive: $\mathcal{O}(|\mathcal{A}|^3 + |\mathcal{A}|^2 n)$

Updating the Hessian

Computing the Hessian and its inverse naively is expensive: $\mathcal{O}(|\mathcal{A}|^3 + |\mathcal{A}|^2 n)$

Fortunately, we can sweep columns of the Hessian and inverse in our out, yielding complexity

- $\mathcal{O}(|\mathcal{D}|^2 n + n|\mathcal{D}||\mathcal{E}| + |\mathcal{D}|^2|\mathcal{E}| + |\mathcal{D}|^3)$ when augmenting the Hessian and
- $\mathcal{O}(|\mathcal{C}|^3 + |\mathcal{C}|^2|\mathcal{E}| + |\mathcal{C}||\mathcal{E}|^2)$ when reducing it,

where

- $\mathcal{C} = \mathcal{A}_{k-1} \setminus \mathcal{A}_k$ (to-be deactivated)
- $\mathcal{D} = \mathcal{A}_k \setminus \mathcal{A}_{k-1}$ (to-be activated)
- $\mathcal{E} = \mathcal{A}_k \cap \mathcal{A}_{k-1}$ (still activate)

Paper V Extras

How Do We Minimize Over One Cluster?

The optimality condition, using the directional derivative, is

$$\forall \delta \in \{-1, 1\}, \quad G'(z; \delta) \geq 0,$$

with

$$\begin{aligned} G'(z; \delta) &= \delta \sum_{j \in \mathcal{C}_k} X_{:j}^\top (X\beta(z) - y) \\ &\quad + H'(z; \delta). \end{aligned}$$

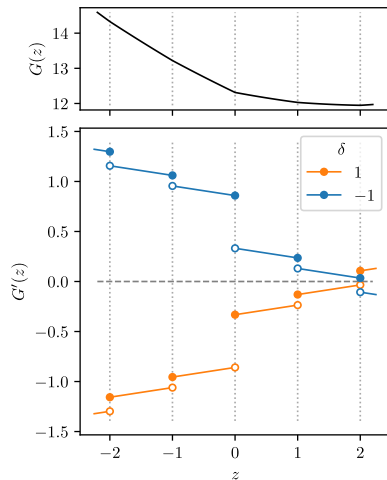


Figure 23: G and its directional derivative $G'(\cdot; \delta)$ for an example with $\beta = [-3, 1, 3, 2]^T$, $k = 1$, and consequently $c^{\setminus k} = \{1, 2\}$.

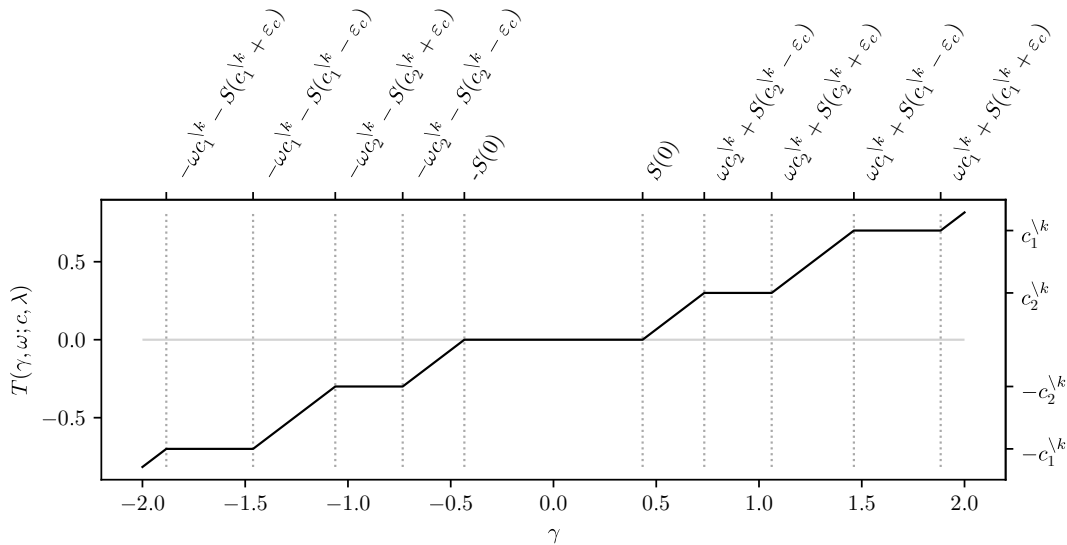


Figure 24: The SLOPE Thresholding Operator

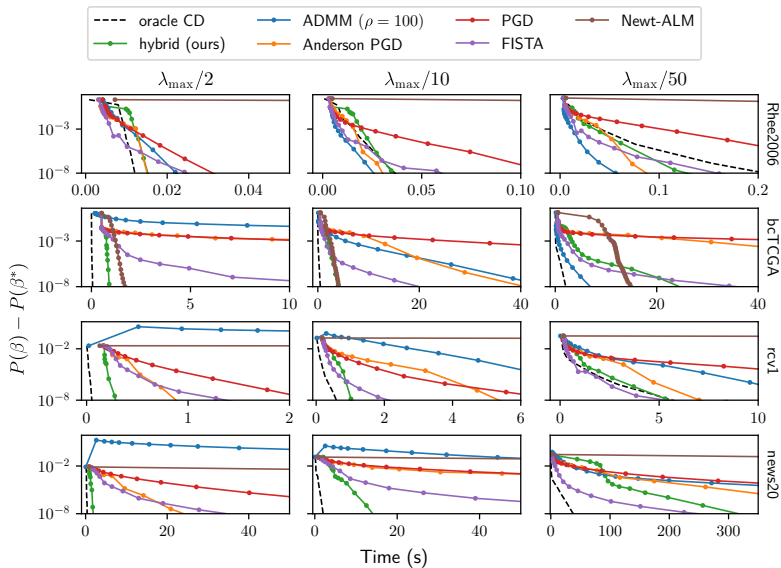


Figure 25: Benchmarks on real data

Paper VI Extras

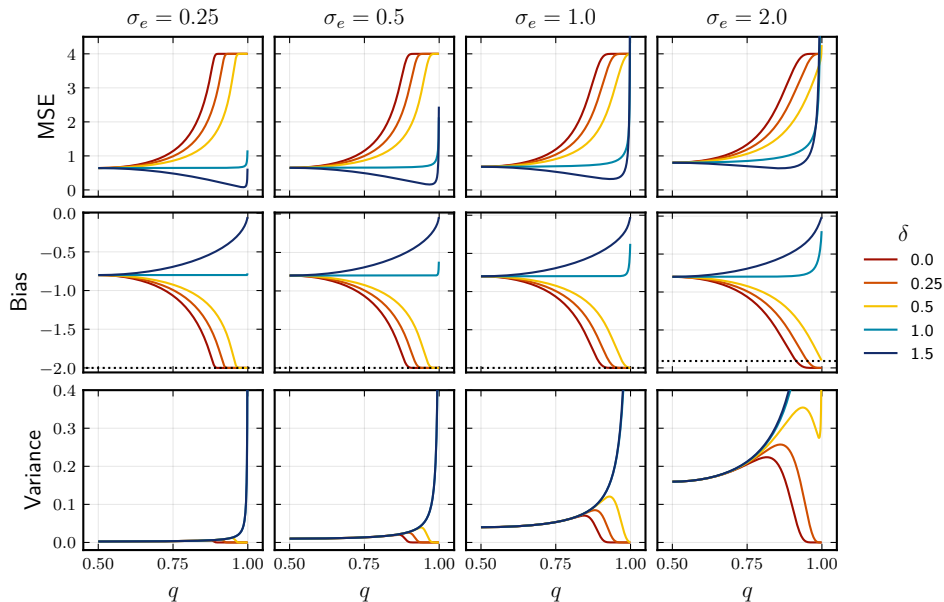
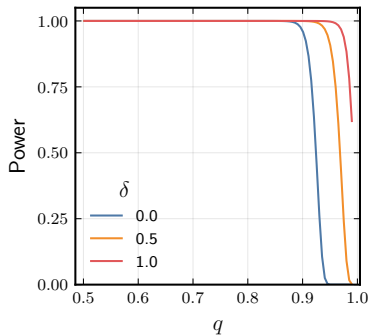


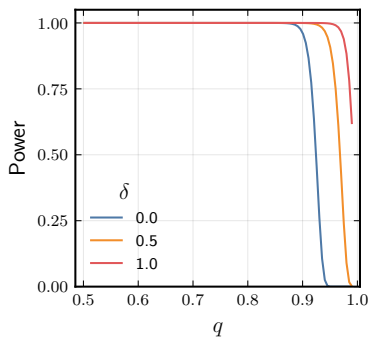
Figure 26: Bias, variance, and mean-squared error for a one-dimensional lasso problem. Theoretical result for orthogonal features. Dotted line is asymptotic result or $\delta = 1/2$.

Lasso example with 10 true signals and varying q and p .

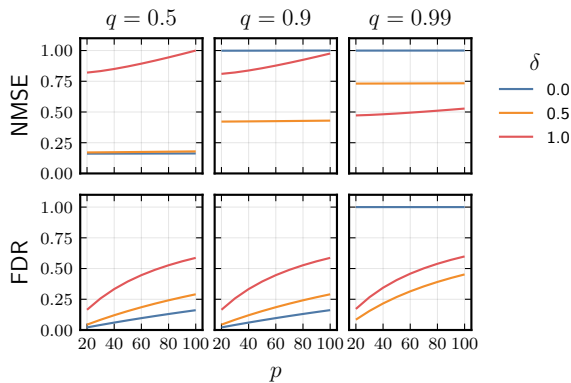


(a) Power in the sense of detecting all the true signals. Constant p .

Lasso example with 10 true signals and varying q and p .



(a) Power in the sense of detecting all the true signals. Constant p .



(b) False discovery rate (FDR) and normalized mean-squared error (NMSE).

Figure 27: Mean squared error (MSE), false discovery rate (FDR), and power.

Theorem

If x_j is a binary feature with class balance $q \in (0, 1)$ and $\lambda_1, \lambda_2 \in (0, \infty)$, $\sigma_\varepsilon > 0$, and $s_j = (q - q^2)^\delta$, $\delta \geq 0$, then

$$\lim_{q \rightarrow 1^+} \mathbb{E} \hat{\beta}_j = \begin{cases} 0 & \text{if } 0 \leq \delta < \frac{1}{2}, \\ \frac{2n\beta_j^*}{n+\lambda_2} \Phi\left(-\frac{\lambda_1}{\sigma_\varepsilon\sqrt{n}}\right) & \text{if } \delta = \frac{1}{2}, \\ \beta_j^* & \text{if } \delta \geq \frac{1}{2}. \end{cases}$$

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and

$$\lim_{q \rightarrow 1^+} \text{Var} \hat{\beta}_j = \begin{cases} 0 & \text{if } 0 \leq \delta < \frac{1}{2}, \\ \infty & \text{if } \delta \geq \frac{1}{2}. \end{cases}$$